LEARNING MODULE

3

Statistical Measures of Asset Returns

by Pamela Peterson Drake, PhD, CFA, and Wu Jian, PhD.

Pamela Peterson Drake, PhD, CFA, is at James Madison University (USA). Jian Wu, PhD, is at State Street (USA).

LEARNING OUTCOMES

Mastery	The candidate should be able to:
☐	calculate, interpret, and evaluate measures of central tendency and location to address an investment problem
☐	calculate, interpret, and evaluate measures of dispersion to address an investment problem
☐	interpret and evaluate measures of skewness and kurtosis to address an investment problem
☐	interpret correlation between two variables to address an investment problem

INTRODUCTION

1

Data have always been a key input for securities analysis and investment management, allowing investors to explore and exploit an abundance of information for their investment strategies. While this data-rich environment offers potentially tremendous opportunities for investors, turning data into useful information is not so straightforward.

This module provides a foundation for understanding important concepts that are an indispensable part of the analytical tool kit needed by investment practitioners, from junior analysts to senior portfolio managers. These basic concepts pave the way for more sophisticated tools that will be developed as the quantitative methods topic unfolds, which are integral to gaining competencies in the investment management techniques and asset classes that are presented later in the CFA curriculum.

This learning module focuses on how to summarize and analyze important aspects of financial returns, including key measures of central tendency, dispersion, and the shape of return distributions—specifically, skewness and kurtosis. The learning module finishes with a graphical introduction to covariance and correlation between two variables, a key concept in constructing investment portfolios to achieve diversification across assets within a portfolio.